

(IaaS) clouds in service-provider environments (the public cloud) or enterprise environments (the private cloud). It supports hypervisors such as VMware vSphere, Oracle, KVM and XenServer for virtualisation. CloudStack implements the Amazon EC2 APIs, Amazon S3 APIs and vCloud API in addition to its own API. CloudStack was originally developed by Cloud.com, which released most of CloudStack under the GNU General Public License, version 3 (GPLv3). Citrix released the remaining code under GPLv3. CloudStack 3.0.2 is the latest stable version, which was released on May 17, 2012. Citrix donated CloudStack to the Apache Software Foundation, and it was accepted as Apache Incubator. Citrix changed the licence to the Apache License version 2.

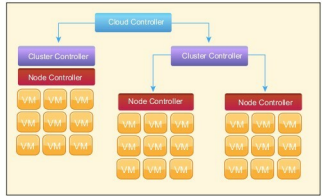


Figure 1: Eucalyptus components

	OpenStack	Eucalyptus	CloudStack
Open Source	Yes	Yes	Yes
Components	Nova (compute) Swift (object storage) Glance (image service) Keystone (identity management) Horizon (GUI interface)	Cloud Controller (CLC): Manages the virtualisation resources and APIs. Provides Web interface. Walrus (S3 storage) Cluster Controller (CC): Controls execution of VMs and their networking. Storage Controller (SC): Provides block-level storage to VMs (EBS). Node Controller (NC): Controls VMs via hypervisors	Management Server. Hypervisor Nodes. Storage Nodes. Layers: Zone, Pod, Cluster, Host, Primary Storage, Secondary Storage
Code-base	Python	Java, C	Java
Hypervisors	Xen, KVM, UML, LXC, VMware	Xen, KVM, VMware	Xen, KVM, VMware, Citrix XenServer
Strengths	Growing community. Corporate support. Single code-base.	Good commercial support. Fault-tolerance. Offers a hybrid-cloud solution with AWS.	Well-rounded GUI Stack is fairly simple. Customisation of the storage back-end.
RightScale Support	Yes	Yes	Yes
API	Yes	Yes	Yes
AWS API Compatibility	Yes	Yes	Yes
Live Migration	Yes	No	Yes
VMware Hypervisor Support (ESX, ESXi, vSphere)	Yes	Yes (Available with subscription to Support)	ESXi
Snapshots	Yes	Yes	Yes
Self-service User Portal	Yes	Yes (Available with subscription to Support)	Yes